BCC1313 – ALGORITHMS & COMPLEXITY

Lab 3

Instructions:

Answer ALL questions.

Question 1

Write a C++/Java program to perform the following operations on a singly linked list:

- Insert a node at the beginning
- Insert a node at the end
- Insert a node at a specific position
- Delete a node at the beginning
- Delete a node at a specific position
- Display the linked list

Hints:

1. Use pointers to traverse the linked list.
2. Handle edge cases such as empty lists and invalid positions.

Question 2

Implement a doubly linked list in C++ that supports the following operations:

- Insert a node at the front
- Insert a node at the end
- Delete a node from the front
- Delete a node from the end
- Traverse the list forwards and backwards

Question 3

You are tasked with designing a music playlist using a singly linked list. Each song has a title and a duration.

- Write a function to add a new song to the playlist.
- Write a function to delete a song from the playlist by its title.
- Write a function to display the entire playlist.

Challenge: Implement a "Next" and "Previous" feature to navigate through the songs.